FINAL PACKAGE

LEGAL OOD PROTOCOL

SUBMISSION_1

Train on 12 families, test on 3 unseen families, ship official-format GPT outputs.

0.722	**0.999**	**0.847**	**15**
OOD3 Top-1	**OOD3 Top-5**	**OOD3 MRR**	**Families**
family-wise avg	family-wise avg	family-wise avg	12 train / 3 test

Submission surface

- nextstep.csv
- completion.csv
- anomaly.csv
- outputs/gpt_large_train12
- n-gram fallback

Score exactly where the jury scores

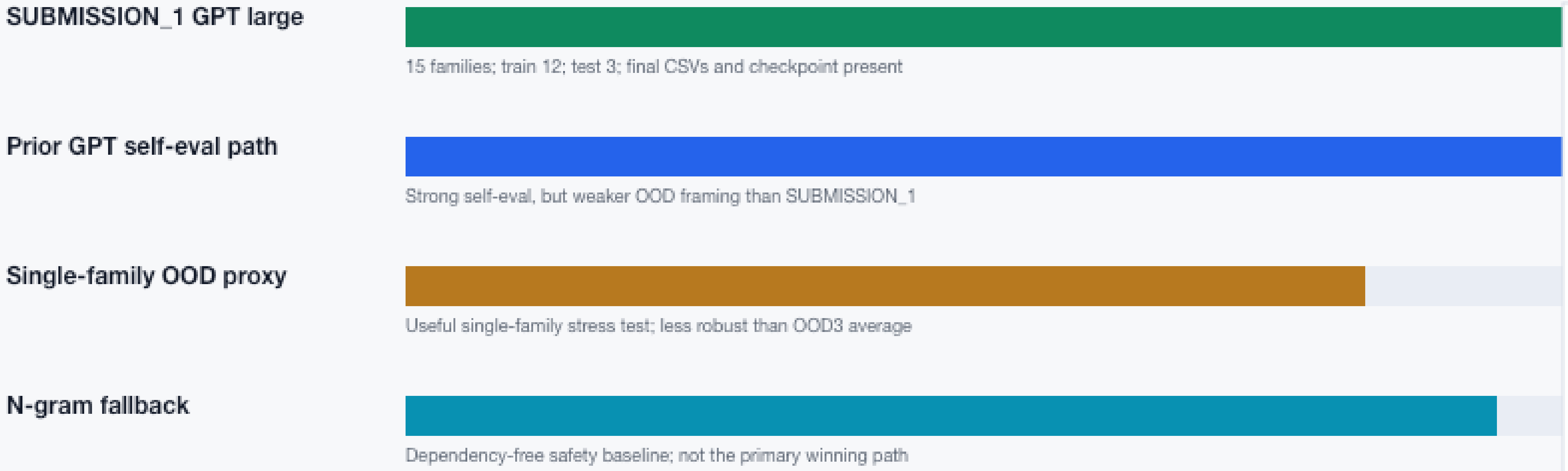
Each official task has a generated CSV, a legal source path, and an auditable local proof.

Jury task	Submitted file	What it scores	SUBMISSION_1 proof	Risk
1. Next step	nextstep.csv	Top-1 / Top-3 / Top-5 / MRR	GPT large OOD3 Top-1 0.722	Low
2. Completion	completion.csv	Normalized edit distance plus exact/token/block	Same checkpoint, official IDs	Medium
3. Anomaly	anomaly.csv	Validity, score, violated rule name	Rule validator, official rule strings	Low
Legality	repo evidence	No hidden eval training	Official eval only used for inference	Low
Reproducibility	REPORT + README	Run path and dependencies	Checkpoint + n-gram fallback	Low

Decision: freeze the number-improvement path here. Polish presentation, not model behavior, unless a format bug is found.

Why this branch ranks first

The best winning chance is the strongest legal OOD result that also ships clean official-format artifacts.



Selection logic

- Higher OOD confidence than inherited single-family proxy
- No eval_input_anomaly training leakage
- Official IDs generated from participant files
- Checkpoint included; fallback remains runnable

Counterargument handled: if the hidden OOD family differs from our synthetic families, the 3-family average is still a better proxy than optimizing on known-format self-eval alone.

OOD protocol is explicit and judge-aligned

SUBMISSION_1 tests whether the model learned process grammar, not just known family routes.

15 total families

12 train families
mosfet, igbt, ic
+ scfam01-09

3 held-out OOD families
scfam10, scfam11, scfam12

Report
family-wise
average

Why this matters

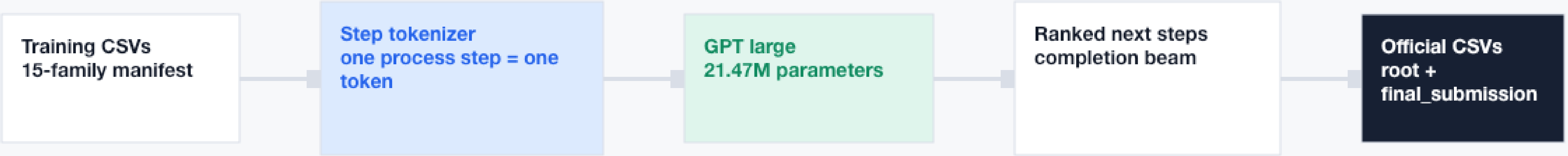
A hidden fourth family rewards transfer. The local score must therefore penalize family memorization before the final submission is frozen.

Legal boundary

Official eval_input_valid.csv and eval_input_anomaly.csv are used only to produce final CSVs, not as training data.

System design stays simple enough to trust

The model improves sequence predictions; the validator owns hard rule compliance.



Fallback path

Trigram n-gram runs without torch and can regenerate official-shaped CSVs from a clean checkout.

Safety path

GPT runner refuses random-init submissions unless explicitly overridden, preventing silent bad CSVs.

Evidence: GPT beats the safe baseline on OOD3

The measured lift is on the train-12/test-3 protocol, not on the final hidden eval files.

+0.058 Top-1 lift vs n-gram

Family-wise OOD3 average: Top-1 0.7216, Top-5 0.9989, MRR 0.8474

Blockers closed before final polish

The package now fails closed on format, legality, and checkpoint availability.

OK **Official IDs**
Root CSVs use valid_0001... and anomaly_... copied through from participant inputs.

OK **Checkpoint present**
outputs/gpt_large_train12 contains config, generation config, and model.safetensors.

OK **Required docs**
REPORT.md, README.md, LICENSE, requirements.txt are present at repo root.

OK **Legal usage**
Official eval inputs are used only for final inference; not for training.

OK **Fallback**
n-gram official bundle remains available if checkpoint transfer fails.

Final files

nextstep.csv

completion.csv

anomaly.csv

Two-minute demo: show trust first, score second

The pitch should make judges comfortable that the result is legal, reproducible, and OOD-aware.

0:00-0:20

Open with the final package

Show root CSVs, REPORT.md, and checkpoint path.

0:20-0:45

Explain train-12/test-3

Point to SUBMISSION_1 manifest and family-wise OOD average.

0:45-1:15

Compare baseline vs GPT

Show n-gram Top-1 0.662 vs GPT Top-1 0.722 on OOD3.

1:15-1:40

Show anomaly safety

Explain rule validator and official rule strings.

1:40-2:00

Close on submission readiness

Official IDs, final CSVs, fallback, no eval training leakage.

Winning claim: SUBMISSION_1 is the best legal bet because it optimizes for hidden-family generalization and ships a complete, reproducible final package.